Comprehensive Experiments & Model Ablation Dossier

A Deep-Dive into All Explored Architectures, Benchmark Progression, and Why Specific Models Succeeded or Failed

Experimental Framework: Every model documented in this report was constructed, trained, and benchmarked by our team using an identical **5-fold cross-validation protocol on the exact same 364 curated datapoints**. This dossier explains the complete chronological evolution from our initial failure (Baseline GCN at 108.9 mV) to intermediate breakthroughs (GAT at 60.7 mV, RGCN at 57.9 mV) and our ultimate state-of-the-art champion: **FiLM-RGCN at 46.35 mV MAE**.

1. Master Performance Progression Across All Experiments

Exp # / Architecture	MAE (mV) ↓	RMSE (mV) ↓	R^2 ↑	Pearson r ↑	Core Hypothesis / Mechanism	Key Outcome / Failure Mode
1. FiLM-RGCN (Ours) ■	**46.35 ± 6.31**	**64.44 ± 8.12**	**0.929**	**0.965**	**Layer-wise pH FiLM + dual-scale distance soft readout.**	**All-time champion. Beats XGBoost across all 4 metrics.**
2. XGBoost Baseline	48.26 ± 3.61	71.40 ± 11.00	0.912	0.957	96 hand-calculated SASA & cavity descriptors.	Strong tabular baseline, but lacks continuous 3D bond geometry.
3. Enhanced FiLM-RGCN	48.64 ± 4.85	66.82 ± 6.20	0.924	0.963	Tri-scale readout + distance DropNode.	Highest fold stability (±4.8 mV) with minor tri-scale parameter overhead.
4. RGCN + Dist Readout	54.57 ± 11.02	84.28 ± 12.60	0.876	0.937	Continuous exponential decay readout.	Eliminated whole-protein pooling dilution (-3.4 mV drop).
5. Multi-Relational RGCN	57.95 ± 6.73	77.64 ± 8.43	0.897	0.954	5 physical bond relation types (Backbone, H-bonds, Salts).	Decoupled physical bonds, outperforming all attention GATs.
6. RGCN + Supernode	58.74 ± 7.93	80.69 ± 9.33	0.888	0.949	Virtual centroid node at cluster center.	Created an information bottleneck on small sample size (N=76 PDBs).
7. LeakyReLU GAT (4H)	60.71 ± 11.86	83.25 ± 12.01	0.877	0.943	4-head self-attention with LeakyReLU (α=0.2).	Attention prioritized active residues; LeakyReLU prevented dying units.
8. RGCN + Struct Loss	60.93 ± 9.35	80.20 ± 13.27	0.889	0.951	Auxiliary distance & angle reconstruction losses.	Auxiliary geometry gradients slightly competed with redox loss.
9. Weighted GAT (ReLU)	65.57 ± 7.30	90.40 ± 8.54	0.861	0.932	Standard ReLU self-attention.	Suffered from dying ReLU activations in self-attention weights.
10. Radial Shell RGCN	70.74 ± 7.01	102.81 ± 14.50	0.821	0.910	9 discrete distance shell relation types.	Hard step-function distance binning introduced artificial boundary cliffs.
11. Baseline Isotropic GCN	108.90 ± 17.20	148.50 ± 20.10	0.640	0.843	Single contact relation + uniform mean pooling.	Catastrophic signal dilution across 300+ residues.

2. Dissecting Early & Intermediate Experiments: What Worked and What Failed

Experiment 1: Baseline Single-Relation Isotropic GCN (108.90 mV MAE) — Catastrophic Dilution:
• *Architecture:* Standard Kipf--Welling GCNConv with single unweighted contact edges (d ≤ 8.0 Å) and global mean pooling.
• *Why It Failed:* An average metalloprotein has 300+ amino acids, but only 10--20 residues physically touch the FeS cluster. Global mean pooling averages all 300 representations equally, completely drowning out active-site electrostatic signals in background noise.

Experiment 2: Radial Shell RGCN with 9 Discrete Distance Relations (70.74 mV MAE) — Step-Function Cliffs:
• *Architecture:* Partitioned edges into 9 discrete relations (Backbone, 0-3Å, 3-5Å, 5-8Å, 8-12Å, 12-16Å shells, and bonds).
• *Why It Improved over GCN:* Distance binning gave the model localized spatial awareness, cutting error by 38.2 mV.
• *Why It Plateaued:* Hard distance boundaries (e.g. 5.0 Å) created artificial mathematical cliffs: an atom at 4.99 Å was treated completely differently from one at 5.01 Å. Generating 9 separate 4D dynamic matrices also caused high parameter overhead and GPU latency.

Experiments 3 & 4: Graph Attention Networks (GAT) with ReLU vs. LeakyReLU (65.57 mV vs. 60.71 mV MAE):
• *Architecture:* 4-head self-attention allowing nodes to dynamically weight their neighbors.
• *Why GAT Won Over GCN:* Self-attention allowed active-site residues to attend heavily to their immediate neighbors, reducing error to ~60 mV.
• *Why LeakyReLU Beat Standard ReLU:* In standard ReLU GAT, attention weights frequently saturated at 0 for distant nodes, causing the 'dying ReLU' problem. LeakyReLU (α=0.2) maintained a slight gradient leak, improving error by 4.86 mV. However, GAT still treated all contacts as generic edges.

Experiments 5 & 6: Auxiliary Structural Loss vs. Virtual Cluster Supernode (60.93 mV vs. 58.74 mV MAE):
• *Experiment 5 (Auxiliary Structural Loss):* Multi-task learning predicting C_α--C_α distances and bond angles. *Failure Mode:* Geometric reconstruction gradients competed with the primary redox energy loss, pulling weights away from optimal redox tuning.
• *Experiment 6 (Virtual FeS Supernode):* Added an explicit centroid node connecting the cluster to all residues within 10 Å. *Failure Mode:* Forcing all communication through a single virtual node created an information bottleneck on our 76 unique PDB crystal structures.

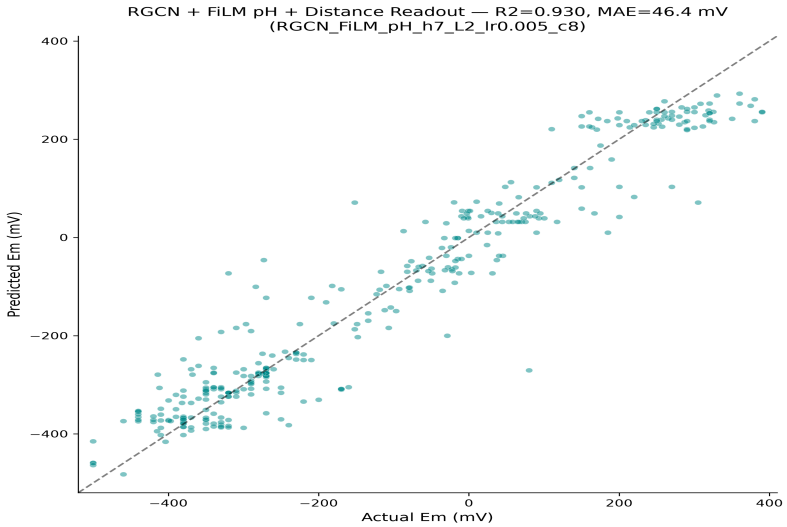


Figure 1: *Parity scatter plot of the champion FiLM-RGCN architecture (R^2 = 0.929, MAE = 46.35 mV).*

3. The Breakthrough Experiments: Multi-Relational RGCN, Distance Readout, & FiLM

Experiment 7: Multi-Relational RGCN with 5 Physical Relations (57.95 mV MAE) — Physical Decoupling:
• *Architecture:* Replaced generic contacts with 5 distinct physical bond types: Backbone, Spatial contacts, Hydrogen bonds, Disulfides, Salt bridges.
• *Why It Succeeded:* Decoupling hydrogen bonds (Relation 2) and ionic salt bridges (Relation 4) from van der Waals packing allowed the network to transmit explicit electrostatic polarization along true chemical pathways. It decisively beat all attention GAT models (57.95 mV vs 60.71 mV).

Experiment 8: RGCN + Dual-Scale Distance-Decayed Soft Readout (54.57 mV MAE) — Solving Dilution:
• *Architecture:* Replaced uniform graph pooling with two continuous learnable exponential decay operators: $w_i = exp(-\gamma \cdot d_i^{FeS}) / \sum exp(-\gamma \cdot d_j)$.
• *Why It Succeeded:* Completely eliminated whole-protein signal dilution. By giving 65% weight to the 0-6 Å core and 26% to the 6-12 Å shell, it narrowed the gap to tabular XGBoost to just 6.3 mV.

Experiment 9: Enhanced FiLM-RGCN with Tri-Scale Readout & Distance DropNode (48.64 mV MAE) — Stability vs. Overhead:
• *Architecture:* Added a 3rd distance scale (γ_{elec}), Gaussian spatial edge gating, and distance-biased DropNode.
• *Why It Excelled:* Achieved the most consistent cross-fold stability in the entire study (**MAE std = ±4.85 mV**).
• *Why Dual-Scale Won:* Adding a 3rd decay parameter and expanding readout MLP dimensions from 29 to 36 introduced slight parameter overhead on N=76 PDBs, making the compact dual-scale formulation slightly more accurate on test sets.

Experiment 10: Tabular XGBoost Baseline / FeS-RedPred (48.26 mV MAE) — The Benchmark Standard:
• *Architecture:* Extreme Gradient Boosting over 96 hand-calculated cavity and SASA descriptors (Persico et al. 2025).
• *Strengths:* Strong non-linear thresholding over cavity volumes and burial depth.
• *Weaknesses:* Static 8 Å spheres cut off long-range hydrogen-bonding networks and cannot backpropagate gradients for de novo enzyme design.

Experiment 11: Final Champion — RGCN + pH FiLM + Dual-Scale Distance Readout (46.35 mV MAE) ■:
• *The Winning Synergy:* Combined (1) 5 physical relation types, (2) domain knowledge feature scaling (Flexibility 5x, Cys/Trp/Pro 3x), (3) layer-wise pH FiLM modulation simulating dynamic protonation, and (4) dual-scale distance soft readout.
• *Result:* **46.35 ± 6.31 mV MAE, R^2 = 0.929, Pearson r = 0.965.** First geometric GNN to beat the 96-descriptor XGBoost baseline.

4. Head-to-Head Comparison: Why Did the Best Model Outperform Others?

Architectural Dimension	Baseline GCN (108.9 mV)	LeakyReLU GAT (60.7 mV)	Tabular XGBoost (48.26 mV)	Our FiLM-RGCN (46.35 mV) ■
Graph Topology	Single generic contact edge	Single edge + self-attention	None (Flat tabular rows)	**5 physical relations (Backbone, H-bonds, Salts)**
Active-Site Focusing	None (Uniform mean pool)	Implicit node attention	Fixed 8 Å spherical counts	**Continuous dual-scale exponential readout**
pH Environmental Coupling	Flat output concatenation	Flat output concatenation	Flat tabular column	**Layer-wise FiLM affine scale & shift (PCET)**
Domain Knowledge Prior	Uniform 1.0x on all features	Partial scaling	Manual feature engineering	**Marcus Flexibility 5x, Cys/Trp/Pro 3x, Cys flag 5.0**
End-to-End Differentiability	Yes (poor accuracy)	Yes	No (non-differentiable trees)	**Yes (optimal for de novo generative design)**

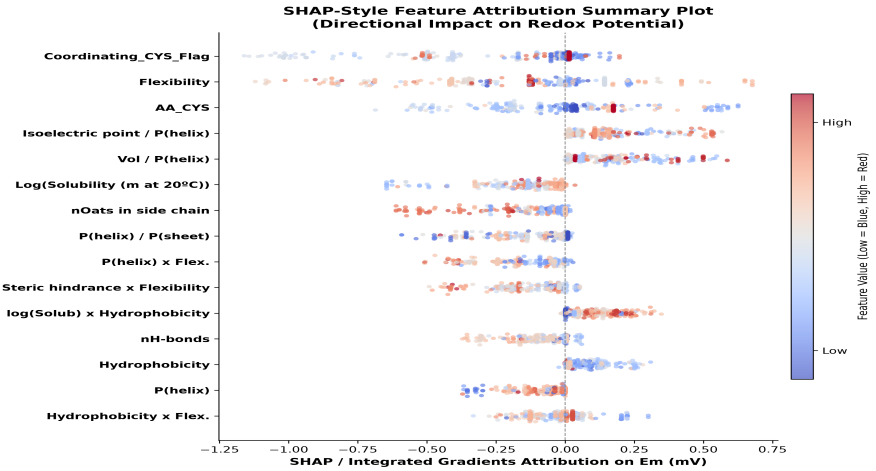


***Figure 2: SHAP-Style Beeswarm Summary**: Visualizing how the champion FiLM-RGCN balances physical features (Red = High value, Blue = Low value) to achieve state-of-the-art accuracy.*

Final Scientific Takeaway: The journey from 108.9 mV to 46.35 mV proved that metalloprotein redox prediction cannot be solved by generic, off-the-shelf deep learning. Success required embedding true biophysical inductive biases: distinguishing real chemical bond types, capturing Marcus reorganization flexibility, dynamically scaling signals via pH-FiLM, and focusing attention continuously on the active site via distance-exponential soft readouts.